

Question

Passing an excess of chlorine gas into a hot concentrated potassium iodide solution, cooling it, and precipitating golden needle-shaped crystals **A** (the reaction forming **A** is a combination reaction).

Heating **A** to constant weight yields a white water-soluble substance **B**, with a weight loss of 75.8%. Collecting all volatile substances from the decomposition products in a cold trap yields orange-yellow crystals **C**, but collecting only the less volatile substances at a relatively high temperature yields a dark red liquid, and further cooling yields dark ruby-colored needle-shaped crystals **D**.

Hint 1: **A** is decomposed by water to give a brown solution, while if **A** is directly decomposed with an excess of base, a colorless solution is obtained. Write the equations for the reactions under both conditions, respectively.

Hint 2: The actual form in which **C** crystals exist is its dimer.

Select the correct option(s) below.

- A.** The anion of **A** and the hybridization of the central atom of molecule **C** are different.
- B.** The anion of **A** is a tetrahedral structure.
- C.** The sum of the reactant coefficients in the ionic equation for the decomposition of **A** in water to obtain a brown solution is 14.
- D.** The sum of the reactant coefficients in the ionic equation for the hydrolysis of **A** to obtain a brown solution is 11.
- E.** The sum of the reactant coefficients in the ionic equation for the decomposition of **A** by excess alkali to obtain a colorless solution is 18.

F. The sum of the product coefficients in the ionic equation for the decomposition of **A** directly with excess base to obtain a colorless solution is 27.

G. In the ionic equation for the decomposition of **A** directly with excess base to obtain a colorless solution, the sum of the reactant coefficients is 21.

H. All of the above statements are incorrect.

Answer

Correct Answer: **C**

Detailed Explanation

Since KI combines with Cl_2 to form **A**, the chemical formula of **A** can be assumed to be KICl_x .

B is most likely KI or KCl

If **B** is KI, then Cl_x is lost,

$35.45x / (39.1 + 35.45x + 126.9) = 75.8\%$, solving for $x = 14.7$, which is discarded

If **B** is KCl, then ICl_{x-1} is lost,

$[126.9 + 35.45(x - 1)] / (39.1 + 35.45x + 126.9) = 75.8\%$

Solving for $x = 4.0$, which meets the requirements

Therefore, **A** is KICl_4 , **B** is KCl

CHECKPOINT

2 PTS

A is KICl_4 , **B** is KCl

Wherein, in **A**, I is sp^3d^2 hybridized, and the ion configuration is planar.

CHECKPOINT

0.5 PTS

In **A**, I is sp^3d^2 hybridized, and the ion configuration is planar

The weight loss is ICl_3 , so **C** is ICl_3 or written as I_2Cl_6 . According to the question, ICl_3 will further decompose when the temperature rises, so **D** is ICl .

CHECKPOINT

2 PTS

C is ICl_3 or written as I_2Cl_6 , **D** is ICl

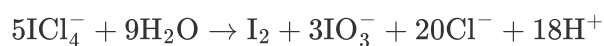
Wherein, in I_2Cl_6 , I is sp^3d^2 hybridized, and two Cl act as bridging ligands, and the molecular configuration is planar.

CHECKPOINT

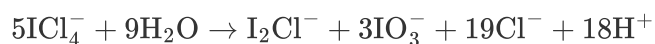
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In I_2Cl_6 , I is sp^3d^2 hybridized, and two Cl act as bridging ligands, and the molecular configuration is planar.

In prompt 1, **A** is decomposed by water to obtain a brown solution. According to the conditions of the question, it is determined that the cause of the brown color is the disproportionation of **A** to generate I_2 . The ion equation of the reaction is as follows:

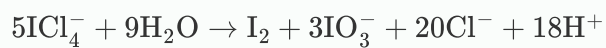


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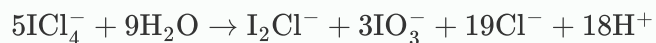


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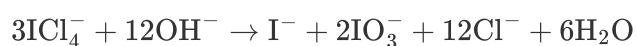
1 PTS



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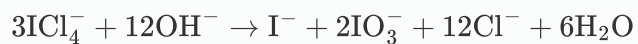


If excess alkali is directly used to decompose **A** to obtain a colorless solution, it directly disproportionates to generate I^- and IO_3^- . The ion equation of the reaction is as follows:



CHECKPOINT

1 PTS



In summary, option **C** is correct.